Large-Signal Stabilization of On-Board Hybrid Power Supply Systems Supplying Constant Power Load Based on Composite Control

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***Abstract*—To meet the cross-timescale response requirements of new electrified loads, lithium batteries and supercapacitors are being integrated into onboard electrical power systems via DC/DC converters. Despite the rapid response and energy recovery capabilities of these storage units, the onboard hybrid power systems remain prone to significant power quality degradation and instability under transient conditions such as wide-range load transients, partial source failures, and pulsed loads with a high peak-to-average ratio. To address these issues, this article proposes a composite controller that integrates a fixed-time sliding mode disturbance observer (FTSMDO) with a prescribed performance controller (PPC). The FTSMDO is designed to estimate lumped disturbances, upon which a PPC is developed using the backstepping method. This approach enhances the system's capability for rapid voltage recovery under various transient events and ensures large-signal stability. Finally, the effectiveness of the proposed composite control strategy is validated through both simulation and experimental results.**

***Index Terms*—Constant power loads (CPLs), hybrid power supply system (HPSS), large-signal stabilization, more-electric aircraft (MEA).**

I. Introduction

WITH the rapid advancement of more-electric and all-electric aircraft technologies, a growing number of manned and autonomous aerial vehicles (AAVs) are incorporating an increasing proportion of onboard electrical equipment. This new generation of electrified loads often exhibits a high peak-to-average power ratio, leading to frequent and severe fluctuations in the power demand of the aircraft's electrical power system. Traditional power systems, which rely on a single type of power source in a centralized configuration, have become inadequate in effectively managing the highly pulsating nature of these loads. This inadequacy seriously compromises the efficiency, safety, and stability of the onboard power supply. Consequently, a transition is underway from conventional single-source architectures to hybrid power systems that integrate multiple types of power sources [1], [2], [3], such as fuel cells (FCs), lithium batteries (LBs), and supercapacitors (SCs), and interfaced via DC/DC converters.

A critical challenge in more-electric aircraft (MEA) is the stabilization of such hybrid power supply systems (HPSSs). The widespread use of constant power loads (CPLs) introduces negative impedance characteristics, which can lead to oscillatory behavior and significant degradation of system stability. This vulnerability is particularly pronounced during large-signal transients, such as abrupt large load changes (such as during the transition from cruise to maneuvering or from taxiing to takeoff), pulsed power demands, and source failures, often resulting in unacceptable voltage deviations or even system instability. These transient events typically lie beyond the valid operating range of small-signal model-based control methods, limiting their ability to ensure stability across all operational regimes.

To overcome the limitations of small-signal-based controller designs, large-signal stabilization techniques are increasingly applied to power electronics dominated systems. These schemes generally comprise two core elements: a disturbance observer to estimate lumped system uncertainties, and a stabilizing controller designed via Lyapunov theory. Prior work has proposed composite controllers integrating nonlinear disturbance observers with backstepping or sliding mode control for DC–DC converters feeding CPLs in DC microgrids [4], [5], [6], [7], [8], [9]. However, these approaches often focus on steady-state or load-step responses and fail to address distinctive onboard transient events such as partial source failure, maneuver-induced power surges, or pulsed loads. Furthermore,

Received 15 December 2025; revised 27 March 2026; accepted 29 April 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62573066 and in part by the Natural Science Foundation of Jiangsu Province under Grant BK20231446. *(Corresponding author: Pengwei Chen.)*

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Digital Object Identifier 10.1109/TIE.2026.3692833

the convergence rate of conventional nonlinear disturbance observers depends on initial conditions, which can degrade the overall response speed and power quality under large disturbances. To achieve convergence independent of initial states, fixed-time disturbance observers (FTDOs) have been introduced for DC–DC converter systems [10], [11], [12], [13], [14]. While these methods ensure state convergence within a fixed time bound, they do not guarantee specific transient performance metrics, such as overshoot and settling time, potentially allowing critical parameters to exceed safe limits during transients.

Prescribed performance control (PPC) offers a systematic way to enforce strict transient and steady-state performance bounds on system responses. For instance, a dynamic PPC scheme based on a generalized proportional-integral observer was proposed in [15] for a buck converter, effectively handling disturbances while avoiding singularity issues. However, this method lacks a fixed-time convergence guarantee, leaving room for improvement in robustness and response speed under extreme conditions. Another study [16] integrated a nonlinear disturbance observer with PPC for a boost converter, confining the output voltage error within prescribed bounds. Yet, the convergence remains asymptotic without a deterministic, initial-condition-independent time bound, making it difficult to meet the stringent response requirements of onboard HPSSs under several transient conditions. Moreover, these studies focus on single-converter systems and do not address the multi-source coordination and global energy management challenges inherent in onboard HPSSs.

To address the above issues, this article proposes a composite controller that combines a fixed-time sliding mode disturbance observer (FTSMDO) with a PPC for onboard HPSSs. The main contributions of this article are summarized as follows.

1) An FTSMDO is designed for the DC/DC converters in FC-LB-SC-ACC HPSS, capable of estimating lumped disturbances from multiple sources-load interactions within a fixed time independent of initial conditions.
2) A PPC is integrated with the FTSMDO to form a composite control scheme, which ensures both fixed-time stability and guaranteed transient/steady-state performance of the DC bus voltage under abrupt load changes and source failures.

The remainder of this article is organized as follows. Section II introduces the architecture of the onboard HPSS and the characteristics of load power consumption. Section III details the design of the proposed controller and its large-signal stability analysis. Sections IV and V present the simulation and experimental results along with the corresponding analysis, respectively. Finally, Section VI concludes the article.

II. System Description

Fig. 1 depicts the architecture of the HPSS studied in this research. Within this architecture, the FC is connected to the DC bus via a boost converter, the LB and SC are interfaced with the DC bus through a bidirectional boost converter, and the energy storage capacitor is connected to the DC bus via a bidirectional buck converter, thereby forming an active capacitor converter (ACC). In Fig. 1, i_{fc}, i_{lb}, i_{sc}, i_{acc}, i_{ofc}, i_{olb}, i_{osc}, i_{oacc}, v_{ofc}, v_{olb}, v_{osc}, and v_{oacc} are the inductor currents, output currents, and voltages of FC, LB, SC, and ACC, respectively. L_{fc}, L_{lb}, L_{sc}, L_{acc}, C_{fc}, C_{lb}, C_{sc}, and C_{acc} are the filter inductors and capacitors of FC, LB, SC, and ACC.

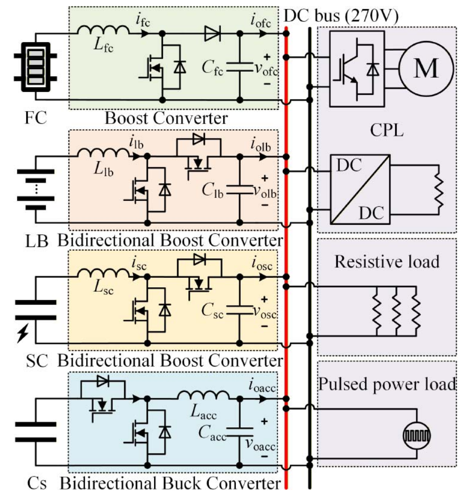

Fig. 1. Architecture of the onboard HPSS.

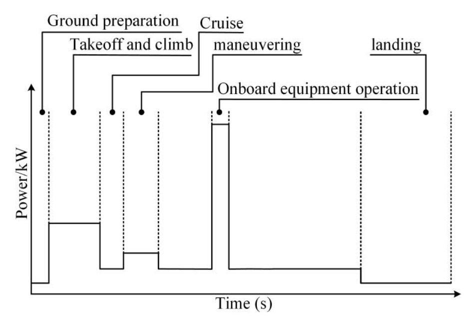

Fig. 2. Schematic diagram of the full-time electrical power characteristics of an AAV.

A schematic diagram of the full-time electrical power characteristics of an AAV during mission execution is depicted in Fig. 2. The profile primarily comprises six phases: ground preparation, takeoff and climb, cruise, maneuvering, onboard equipment operation, and landing. As shown in the diagram, the power required during the takeoff and climb phase as well as the maneuvering phase is several times higher than that during the cruise phase, while the power demand during the onboard equipment operation phase exceeds tenfold that of the cruise phase. Given the high energy density of FC units, they should provide a power output at least equivalent to the cruise power to meet the long-endurance operational requirements of the AAV. In contrast, the transient high-power output and energy recovery during the takeoff and climb, and maneuvering phases should be supplied by LB units and SC units. The power for onboard equipment operation is entirely provided by ACC.

III. Design of the Proposed Composite Controller

A. Preliminary

The core idea of PPC is to transform performance constraints into control constraints, typically via a performance function, and embed them into the controller design process, thereby ensuring that control metrics such as overshoot, convergence speed, and steady-state error satisfy the prescribed performance standards [17]. The underlying principle is as follows.

The system error is constrained to a prescribed performance envelope via the performance function, which can be expressed as follows:

$$-\underline{\delta}\rho(t) \leq e(t) \leq \overline{\delta}\rho(t), \forall t \geq 0 \tag{1}$$

where $\underline{\delta}$ and $\overline{\delta}$ are design parameters, $e(t)$ denotes the state error, and $\rho(t)$ represents the performance function. A commonly used form of the performance function is the exponentially decaying type, which is expressed as follows:

$$\rho(t) = (\rho(0) - \rho(\infty))\, e^{-gt} + \rho(\infty) \tag{2}$$

where $\rho(0)$ denotes the initial permissible error bound, $\rho(\infty)$ represents the steady-state permissible error bound, and g is the convergence rate parameter. Through this function, the transient and steady-state performance of the error can be prescribed.

As inferred from (1) and (2), the error boundary constraint is nonlinear. Direct controller design consequently involves intricate conditional logic. Therefore, a nonlinear mapping is utilized to convert the constrained control problem into an unconstrained stability problem, outlined as follows.

First, the error $e(t)$ is converted to a relative error with respect to a reference

$$q(t) = \frac{e(t)}{\rho(t)}. \tag{3}$$

Then,

$$-\underline{\delta} \leq q(t) \leq \overline{\delta}. \tag{4}$$

Subsequently, the error is mapped into an unconstrained space via a strictly monotonic invertible function

$$\xi = \Gamma(q(t)) = \frac{1}{2}\ln\frac{\underline{\delta} + q(t)}{\overline{\delta} - q(t)}. \tag{5}$$

The derivative of (5) is given by

$$\dot{\xi} = k_\xi\left(\dot{e}(t) - \frac{\dot{\rho}(t)}{\rho(t)}e(t)\right) \tag{6}$$

where $k_\xi = (\overline{\delta} + \underline{\delta})/2(\underline{\delta} + q(t))(\overline{\delta} - q(t))\rho(t)$.

The design of the controller ensures that as $\xi(t) \to 0$, the incorporation of the performance constraint condition is achieved.

B. Design of the Power Sharing Controller

The analysis in our prior work [18] dictates that the power sharing (excluding pulsed power loads) across the FC, LB, and SC must match their dynamic response capabilities, i.e., the FC for low-frequency, the LB for mid-frequency, and the SC for high-frequency power. Given this, the power-sharing control can be implemented via a decentralized strategy utilizing virtual impedance droop control [18], [19], [20], formulated as

$$v_{\mathrm{ofc}} = v_{\mathrm{nom}} - \frac{R_{\mathrm{vfc}}L_{\mathrm{vfc}}s}{L_{\mathrm{vfc}}s + R_{\mathrm{vfc}}}i_{\mathrm{ofc}} \tag{7}$$

$$v_{\mathrm{olb}} = v_{\mathrm{nom}} - R_{\mathrm{vlb}}i_{\mathrm{olb}} \tag{8}$$

$$v_{\mathrm{osc}} = v_{\mathrm{nom}} - \frac{R_{\mathrm{vsc}}}{R_{\mathrm{vsc}}C_{\mathrm{vsc}}s + 1}i_{\mathrm{osc}} \tag{9}$$

where v_{nom} is the nominal voltage of the DC bus, R_{vfc}, L_{vfc}, R_{vlb}, R_{vsc}, and C_{vsc} are the virtual impedance droop coefficients of FC converter, LB converter, and SC converter, respectively.

Owing to the high peak-to-average ratio and transient nature of the pulsed load, employing virtual impedance droop for power sharing yields poor performance. Accordingly, the control scheme was retained as the current feedforward approach, following the methodologies presented in [21] and [22].

C. Model Transformation

The boost converter, bidirectional boost converters, and all the loads in Fig. 1 can be modeled as

$$\begin{cases} L_i\dfrac{di_{L_i}}{dt} = v_i - (1-u_i)v_{oi} \\ C_i\dfrac{dv_{oi}}{dt} = (1-u_i)i_{L_i} - \dfrac{v_{oi}}{R} - \dfrac{P}{v_{oi}} \end{cases} \tag{10}$$

where the subscript i can denote FC, LB, or SC. L_i, C_i, v_i, v_{oi}, i_{L_i}, and u_i are the inductors, capacitors, input voltage, output voltage, inductor currents, and duty cycles of the converters, respectively. R represents the lumped resistive load, while P signifies the lumped CPL.

To facilitate the subsequent controller design, (10) should be transformed into the Brunovsky canonical form using the exact feedback linearization technique proposed in [18], which ensures the transformation is a diffeomorphism. The transformation steps are as follows.

First, the total energy in (10) can be expressed as

$$x_{1i} = \frac{L_i i_{L_i}^2}{2} + \frac{C_i v_{oi}^2}{2}. \tag{11}$$

Differentiating (11) yields

$$\dot{x}_{1i} = L_i i_{L_i}\dot{i}_{l_i} + C_i v_{oi}\dot{v}_{oi} = v_i i_{L_i} - \frac{v_{oi^2}}{R} - P. \tag{12}$$

Then, we define the new state x_{2i} and uncertainty d_{1i} from (12) as

$$x_{2i} = v_i i_{L_i} - \frac{v_{oi}^2}{R_0} \tag{13}$$

$$d_{1i} = -P + \frac{v_{oi}^2}{R_0} - \frac{v_{oi}^2}{R} \tag{14}$$

where R_0 denotes the nominal value of the resistive load.

Differentiating (13) yields

$$\dot{x}_{2i} = \frac{v_i^2}{L_i} + \frac{2v_{oi}^2}{R_0^2} - \left(\frac{v_i v_{oi}}{L_i} + \frac{2i_{L_i}v_{oi}}{R_0^2 C_i}\right)(1-u_i) + \frac{2}{R_0 C_i}\left(P - \frac{v_{oi}^2}{R_0} + \frac{v_{oi}^2}{R}\right). \tag{15}$$

From (15), we define the intermediate control law ϑ_i and the system uncertainty d_{2i} as

$$\begin{cases}\vartheta_i = \dfrac{v_i^2}{L_i} + \dfrac{2v_{oi}^2}{R_0^2} - \left(\dfrac{v_i v_{oi}}{L_i} + \dfrac{2i_{L_i} v_{oi}}{R_0^2 C_i}\right)(1-u_i) \\ d_2 = \dfrac{2}{R_0 C_i}\left(P - \dfrac{v_{oi}^2}{R_0} + \dfrac{v_{oi}^2}{R}\right).\end{cases} \tag{16}$$

Finally, the Brunovsky canonical form of (10) can be expressed as

$$\begin{cases}\dot{x}_{1i} = x_{2i} + d_{1i} \\ \dot{x}_{2i} = \vartheta_i + d_{2i}.\end{cases} \tag{17}$$

D. Design of the FTSMDO

It can be observed from (17) that the system uncertainties d_{1i} and d_{2i} present a major obstacle for controller design. Hence, an FTSMDO is adopted to observe these uncertainties, aimed at facilitating the further design process.

Building on the work in [14], [23], and [24], an FTSMDO for the uncertainties d_{1i} and d_{2i} in (17) can be constructed as

$$\begin{cases}\dot{\hat{x}}_{1i} = \hat{d}_{1i} + x_{2i} + k_{11i}\mathrm{sig}^{\alpha_{1i}}(z_{1i}) + k_{12i}\mathrm{sig}^{\beta_{1i}}(z_{1i}) \\ \dot{\hat{d}}_{1i} = \hat{\dot{d}}_{1i} + k_{21i}\mathrm{sig}^{\alpha_{2i}}(z_{1i}) + k_{22i}\mathrm{sig}^{\beta_{2i}}(z_{1i}) \\ \dot{\hat{\dot{d}}}_{1i} = k_{31i}\mathrm{sig}^{\alpha_{3i}}(z_{1i}) + k_{32i}\mathrm{sig}^{\beta_{3i}}(z_{1i})\end{cases} \tag{18}$$

where $z_{1i} = x_{1i} - \hat{x}_{1i}$, $\mathrm{sig}^o(\cdot)$ denotes $|\cdot|^o\mathrm{sign}(\cdot)$, $\alpha_{1i}, \alpha_{2i}, \alpha_{3i}, \beta_{1i}, \beta_{2i}, \beta_{3i}, k_{11i}, k_{12i}, k_{21i}, k_{22i}, k_{31i}$, and k_{32i} are system parameters and must satisfy the following conditions.

1) $\alpha_{ji} = j\alpha - (j-1)$, $\alpha \in (1-\varepsilon_1, 1)$, $\beta_{ji} = j\beta - (j-1)$, $\beta \in (1, 1+\varepsilon_2)$ $(j = 1,2,3)$, ε_1 and ε_2 are small positive real numbers.
2) The coefficient k_{11i}, k_{12i}, k_{21i}, k_{22i}, k_{31i}, and k_{32i} is required for the following matrix to be Hurwitz:

$$A_i = \begin{bmatrix}-k_{11i} & 1 & 0 \\ -k_{21i} & 0 & 1 \\ -k_{31i} & 0 & 0\end{bmatrix}, \begin{bmatrix}-k_{12i} & 1 & 0 \\ -k_{22i} & 0 & 1 \\ -k_{32i} & 0 & 0\end{bmatrix}.$$

After applying the aforementioned FTSMDO and assuming that the disturbances d_{1i} and d_{2i} are differentiable with $|\dot{d}_{1i}| \le \varsigma_1$ and $|\dot{d}_{2i}| \le \varsigma_2$ (ς_1 and ς_1 are small enough constant), it follows from the results in [24] that the upper bound for the convergence time of the disturbance estimation error will be T_{oi}. And according to (14) and (16), $\hat{d}_{2i}$ can be directly derived as

$$\hat{d}_{2i} = -\frac{2}{R_0 C_i}\hat{d}_{1i}. \tag{19}$$

E. Design of the PPC

For the HPSS studied in this article, which utilizes virtual impedance droop control for dynamic power sharing, the control objective of the converters is to rapidly track the reference voltage generated by the droop controller. Therefore, for system (17), the tracking target should be

$$\begin{aligned}T_i &= x_{1i}^* \\ &= \frac{1}{2}L_i i_{L\mathrm{iref}}^2 + \frac{1}{2}C_i v_{o\mathrm{iref}}^2 \\ &= \frac{1}{2}L_i\left(\frac{P_{\mathrm{ref}}}{v_i}\right)^2 + \frac{1}{2}C_i v_{o\mathrm{iref}}^2 \\ &= \frac{1}{2}L_i\left(\frac{\frac{v_{o\mathrm{iref}}^2}{R_0} - \hat{d}_{1i}}{v_i}\right)^2 + \frac{1}{2}C_i v_{o\mathrm{iref}}^2.\end{aligned} \tag{20}$$

To facilitate the design of a PPC via the backstepping method, the following new coordinates are defined:

$$\begin{cases}e_{1i} = x_{1i} - x_1^* \\ e_{2i} = x_{2i} - x_2^*.\end{cases} \tag{21}$$

Since the control objective is to track the reference voltage, and its performance is primarily evaluated based on the dynamic and steady-state performance of the voltage, the tracking error e_{1i} of the state variable x_{1i} associated with the reference voltage is incorporated into the unconstrained function ξ_i of the performance function, which can be expressed as

$$\xi_i = \frac{1}{2}\ln\frac{\underline{\delta} + e_{1i}/\rho(t)}{\bar{\delta} - e_{1i}/\rho(t)}. \tag{22}$$

Differentiating (22) yields

$$\dot{\xi}_i = k_{\xi_i}\left(x_{2i} + d_{1i} - \dot{x}_1^* - \frac{\dot{\rho}_i(t)}{\rho_i(t)}e_{1i}\right). \tag{23}$$

To stabilize (23) in a fixed time, the virtual control law can be designed as

$$x_{2i}^* = -\frac{\zeta_{1i}}{k_{\xi_i}}\varphi(\xi_i) - \hat{d}_{1i} + \dot{x}_{1i}^* + \frac{\dot{\rho}_i(t)}{\rho_i(t)}e_{1i} \tag{24}$$

where

$$\varphi_{1i}(\xi_i) = a_{1i}\mathrm{sig}(\xi_i)^{\frac{m_{1i}}{n_{1i}}} + b_{1i}\mathrm{sig}(\xi_i)^{\frac{p_{1i}}{q_{1i}}} \tag{25}$$

where $a_{1i} > b_{1i} > 0$, $m_{1i} > n_{1i} > 0$, $q_{1i} > p_{1i} > 0$, and they are odd integers number, $\zeta_{1i} > 0$.

Next, the derivative of e_{2i} can be expressed as

$$\dot{e}_{2i} = \vartheta_i + d_{2i} - \dot{x}_2^*. \tag{26}$$

Similarly, to stabilize (26) in a fixed time, the intermediate control law can be designed as

$$\vartheta_i = -\zeta_{2i}\varphi_2(e_{2i}) - \hat{d}_{2i} + \dot{x}_{2i}^* \tag{27}$$

where

$$\varphi_{2i}(e_{2i}) = a_{2i}\mathrm{sig}(e_{2i})^{\frac{m_{2i}}{n_{2i}}} + b_{2i}\mathrm{sig}(e_{2i})^{\frac{p_{2i}}{q_{2i}}} \tag{28}$$

where $a_{2i} > b_{2i} > 0$, $m_{2i} > n_{2i} > 0$, $q_{2i} > p_{2i} > 0$, and they are odd integers number, $\zeta_{2i} > 0$.

Finally, according to (16), the final control law for (10) can be expressed as

$$u_i = 1 - \left(\frac{v_i^2}{L_i} + \frac{2v_{oi}^2}{R_0^2 C_i} - \vartheta_i\right) \Big/ \left(\frac{v_i v_{oi}}{L_i} + \frac{2i_{Li}v_{oi}}{R_0 C_i}\right). \tag{29}$$

F. Large-Signal Stability Analysis

Theorem 1: After adopting the FTSMDO (18) and the PPC (22), (24), and (27), systems (10) and (17) are large-signal stable, and the output voltage v_{oi} can converge to the reference value v_{oiref} in fixed time with an upper bound of the convergence time being T_{total}.

Proof: Since both the controller and disturbance observer are designed to be finite-time convergent, the large-signal stability analysis of the system is conducted in two stages.

Stage 1: In this stage, the FTSMDO has not yet converged ($t < T_{oi}$), and the Lyapunov function is considered as

$$V = \frac{1}{2}\xi_i^2 + \frac{1}{2}e_{2i}^2. \tag{30}$$

At this point, the system state error dynamics can be expressed as

$$\begin{cases} \dot{\xi}_i = k_{\xi_i}\left(e_{2i} - \dfrac{\zeta_{1i}}{k_{\xi_i}}\varphi(\xi_i) + \tilde{d}_{1i}\right) \\ \dot{e}_{2i} = -\zeta_{2i}\varphi(e_{2i}) - \dfrac{2}{R_0C}\tilde{d}_{1i}. \end{cases} \tag{31}$$

Then, the derivative of V can be expressed as

$$\begin{aligned} \dot{V} &= \xi_i\dot{\xi}_i + e_{2i}\dot{e}_{2i} \\ &= \xi_i k_{\xi_i}(e_{2i} - \frac{\zeta_{1i}}{k_{\xi_i}}\varphi(\xi_i) + \tilde{d}_{1i}) \\ &\quad + e_{2i}(-\zeta_{2i}\varphi(e_{2i}) - \frac{2}{R_0C}\tilde{d}_{1i}) \\ &\le k_{\xi_i}\xi_i e_{2i} - \zeta_{1i}a_{1i}\left(\xi_i^2\right)^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \zeta_{1i}b_{1i}\left(\xi_i^2\right)^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} \\ &\quad - \zeta_{2i}a_{2i}\left(e_{2i}^2\right)^{\frac{m_{2i}+n_{2i}}{2n_{2i}}} - \zeta_{2i}b_{2i}\left(e_{2i}^2\right)^{\frac{p_{2i}+q_{2i}}{2q_{2i}}} \\ &\quad + \left(k_{\xi_i}\xi_i - \frac{2}{R_0C_i}e_{2i}\right)\tilde{d}_{1i}. \end{aligned} \tag{32}$$

Utilizing Young's Inequality, it can be inferred that

$$\xi_i k_{\xi_i} e_{2i} \le \frac{1}{2}\bar{k}_{\xi_i}\xi_i^2 + \frac{1}{2}\bar{k}_{\xi_i}e_{2i}^2 \tag{33}$$

$$\begin{aligned} \left(k_{\xi_i}\xi_i - \frac{2}{R_0C_i}e_{2i}\right)\tilde{d}_{1i} &\le \left|k_{\xi_i}\xi_i - \frac{2}{R_0C_i}e_{2i}\right|\left|\tilde{d}_{1i}\right| \\ &\le \left(k_{\xi_i}|\xi_i| + \frac{2}{R_0C_i}|e_{2i}|\right)\left|\tilde{d}_{1i}\right| \\ &\le \frac{1}{2\eta}\left(\bar{k}_{\xi_i}|\xi_i| + \frac{2}{R_0C_i}|e_{2i}|\right)^2 + \frac{\eta}{2}\tilde{d}_{1i}^2 \\ &\le \frac{1}{2\eta}\left(2\bar{k}_{\xi_i}^2\xi_i^2 + \frac{8}{R_0^2C_i^2}e_{2i}^2\right) + \frac{\eta}{2}\tilde{d}_{1i}^2 \end{aligned} \tag{34}$$

where $\bar{k}_{\xi_i}$ denotes the maximum value of k_{ξ_i}, η is a positive number. Substituting (33) and (34) into (32) yields

$$\begin{aligned} \dot{V} &= \frac{1}{2}\left(\bar{k}_{\xi_i} + \frac{2\bar{k}_{\xi_i}^2}{\eta}\right)\xi_i^2 + \frac{1}{2}\left(\bar{k}_{\xi_i} + \frac{8}{\eta R_0^2C_i^2}\right)e_{2i}^2 \\ &\quad - \zeta_{1i}a_{1i}\left(\xi_i^2\right)^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \zeta_{1i}b_{1i}\left(\xi_i^2\right)^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} \\ &\quad - \zeta_{2i}a_{2i}\left(e_{2i}^2\right)^{\frac{m_{2i}+n_{2i}}{2n_{2i}}} - \zeta_{2i}b_{2i}\left(e_{2i}^2\right)^{\frac{p_{2i}+q_{2i}}{2q_{2i}}} \\ &\quad + \frac{\eta}{2}\tilde{d}_{1i}^2. \end{aligned} \tag{35}$$

According to lemma 2 in [14], we have the following:

$$\begin{aligned} \dot{V} &\le \Theta\left(\frac{1}{2}\xi_i^2 + \frac{1}{2}e_{2i}^2\right) \\ &\quad - \kappa_{1i}2^{\frac{m_{1i}+n_{1i}}{2n_{1i}}}2^{\frac{n_{1i}-m_{1i}}{2n_{1i}}}\left(\frac{1}{2}\xi_i^2 + \frac{1}{2}e_{2i}^2\right)^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} \\ &\quad - \kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}\left(\frac{1}{2}\xi_i^2 + \frac{1}{2}e_{2i}^2\right)^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} \\ &\le \Theta V - 2\kappa_{1i}V^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}V^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} + \frac{\eta}{2}\tilde{d}_{1i}^2 \end{aligned} \tag{36}$$

where $\Theta = \max\{\bar{k}_{\xi_i} + 2\bar{k}_{\xi_i}^2/\eta, \bar{k}_{\xi_i} + 8/(\eta R_0^2C_i^2)\}$, $\kappa_{1i} = \min\{\zeta_{1i}a_{1i}, \zeta_{2i}a_{2i}\}$, $\kappa_{2i} = \min\{\zeta_{1i}b_{1i}, \zeta_{2i}b_{2i}\}$.

When $V \ge 1$, $\Theta V \le \Theta V^{(m_{1i}+n_{1i}/2n_{1i})}$, and thus

$$\dot{V} \le -\left(2\kappa_{1i} - \Theta\right)V^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}V^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} + \frac{\eta}{2}\tilde{d}_{1i}^2. \tag{37}$$

According to lemma 2 in [23], this indicates that the state error of the system will converge to the following neighborhood in finite time:

$$\xi_i, e_{2i} \in \left\{\xi_i, e_{2i}\middle| V \le \min\left\{\left(\frac{\eta\tilde{d}_{1i}^2/2}{(1-\mu)(2\kappa_{1i}-\Theta)}\right)^{\frac{2n_{1i}}{m_{1i}+n_{1i}}}, \left(\frac{\eta\tilde{d}_{1i}^2/2}{(1-\mu)\kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}}\right)^{\frac{2q_{1i}}{p_{1i}+q_{1i}}}\right\}\right\}. \tag{38}$$

When $V < 1$, $\Theta V \le \Theta V^{(p_{1i}+q_{1i}/2q_{1i})}$, and thus

$$\dot{V} \le -2\kappa_{1i}V^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \left(\kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} - \Theta\right)V^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} + \frac{\eta}{2}\tilde{d}_{1i}^2. \tag{39}$$

Similarly, the system state error will converge to the following neighborhood in finite time:

$$\xi_i, e_{2i} \in \left\{\xi_i, e_{2i}\middle| V \le \min\left\{\left(\frac{\eta\tilde{d}_{1i}^2/2}{(1-\mu)2\kappa_{1i}}\right)^{\frac{2n_{1i}}{m_{1i}+n_{1i}}}, \left(\frac{\eta\tilde{d}_{1i}^2/2}{(1-\mu)(\kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}-\Theta)}\right)^{\frac{2q_{1i}}{p_{1i}+q_{1i}}}\right\}\right\}. \tag{40}$$

Consequently, the closed-loop system is input-to-state stable over the time interval $[0,\ T_o]$

Stage 2: In this stage ($t > T_o$), based on the results in [24], it can be concluded that the disturbance observer has converged, and thus the system state error dynamics can be reexpressed as

$$\begin{cases} \dot{\xi}_i = k_{\xi_i}\left(e_{2i} - \dfrac{\zeta_{1i}}{k_{\xi_i}}\varphi(\xi_i)\right) \\ \dot{e}_{2i} = -\zeta_{2i}\varphi(e_{2i}). \end{cases} \tag{41}$$

Substituting (41) into $\dot{V}$ yields the following:

$$\begin{aligned}\dot{V} &= k_{\xi_i}\xi_i e_2 - \xi_i\zeta_{1i}\varphi_{1i}(\xi_i) - e_{2i}\zeta_{2i}\varphi_{2i}(e_{2i})\\ &\le k_{\xi_i}\xi_i e_2 - \zeta_{1i}a_{1i}\left(\xi_i^2\right)^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \zeta_{1i}b_{1i}\left(\xi_i^2\right)^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}\\ &\quad - \zeta_{2i}a_{2i}\left(e_{2i}^2\right)^{\frac{m_{2i}+n_{2i}}{2n_{2i}}} - \zeta_{2i}b_{2i}\left(e_{2i}^2\right)^{\frac{p_{2i}+q_{2i}}{2q_{2i}}}. \end{aligned}\tag{42}$$

Utilizing Young's inequality, it can be inferred that

$$\begin{aligned}\dot{V} &\le \bar{k}_{\xi_i}\left(\frac{1}{2}\xi_i^2 + \frac{1}{2}e_{2i}^2\right)\\ &\quad - \kappa_{1i}2^{\frac{m_{1i}+n_{1i}}{2n_{1i}}}2^{\frac{n_{1i}-m_{1i}}{2n_{1i}}}\left(\frac{1}{2}\xi_i^2 + \frac{1}{2}e_{2i}^2\right)^{\frac{m_{1i}+n_{1i}}{2n_{1i}}}\\ &\quad - \kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}\left(\frac{1}{2}\xi_i^2 + \frac{1}{2}e_{2i}^2\right)^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}.\end{aligned}$$

When $V \ge 1$, $\bar{k}_{\xi_i}V \le \bar{k}_{\xi_i}V^{(m_{1i}+n_{1i}/2n_{1i})}$, and thus

$$\begin{aligned}\dot{V} &\le \bar{k}_{\xi_i}V - 2\kappa_{1i}V^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}V^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}\\ &\le -\left(2\kappa_{1i} - \bar{k}_{\xi_i}\right)V^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}V^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}.\end{aligned}\tag{43}$$

When $V < 1$, $\bar{k}_{\xi_i}V \le \bar{k}_{\xi_i}V^{(p_{1i}+q_{1i}/2q_{1i})}$, and thus

$$\begin{aligned}\dot{V} &\le \bar{k}_{\xi_i}V - 2\kappa_{1i}V^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}V^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}\\ &\le -2\kappa_{1i}V^{\frac{m_{1i}+n_{1i}}{2n_{1i}}} - \left(\kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} - \bar{k}_{\xi_i}\right)V^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}.\end{aligned}\tag{44}$$

According to lemma 1 in [23], we can conclude that the system states error will converge to 0 in a fixed time; the upper bound of the convergence time is as follows:

$$\begin{aligned}T_{ci} \le \max\Bigg\{&\frac{1}{2\kappa_{1i} - \bar{k}_{\xi_i}}\frac{2n_{1i}}{m_{1i}-n_{1i}} + \frac{1}{\kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}}}\frac{2q_{1i}}{q_{1i}-p_{1i}},\\ &\frac{1}{2\kappa_{1i}}\frac{2n_{1i}}{m_{1i}-n_{1i}} + \frac{1}{\kappa_{2i}2^{\frac{p_{1i}+q_{1i}}{2q_{1i}}} - \bar{k}_{\xi_i}}\frac{2q_{1i}}{q_{1i}-p_{1i}}\Bigg\}.\end{aligned}\tag{45}$$

This completes the proof.

Based on the above procedure, it can be demonstrated that after selecting appropriate control parameters, the boost/bidirectional boost converters in the HPSS achieve large-signal stability. Its stabilization process is divided into two stages: stage 1 (corresponding to the time interval $(0,\ T_{oi})$) is input-to-state stable, while stage 2 (corresponding to the time interval $(T_{oi},\ T_{oi}+T_{ci})$) is globally asymptotically stable (large-signal stable).

IV. Simulation Results

To verify the effectiveness of the proposed composite controller in Fig. 3, a simulation model based on the system architecture shown in Fig. 1 was first built in Matlab/Simulink, with the circuit parameters listed in Table I. Moreover, to examine the worst case scenario, the resistance R_0 was set to a sufficiently large value in the simulation, such that all loads can be approximately treated as CPLs.

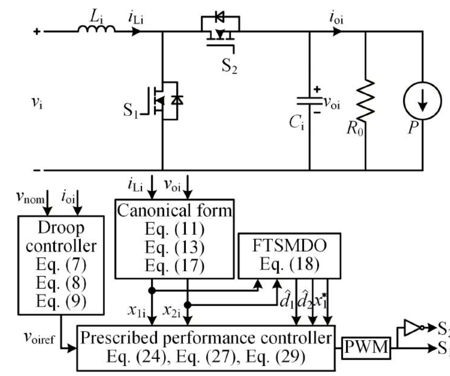

Fig. 3. Simplified schematic of the control strategy.

TABLE I
System Parameters

Symbol	Description	Value
v_{nom}	Nominal voltage of DC bus	270 V
$L_{\text{fc}}, L_{\text{lb}}, L_{\text{sc}}$	Filter inductor of FC/LB/SC converter	0.6 mH
$C_{\text{fc}}, C_{\text{lb}}, C_{\text{sc}}$	Filter capacitor of FC/LB/SC converter	1470 μF
f_{sw}	Switching frequency	20 kHz
R_{vfc}	Resistive droop coefficient of FC converter	2 Ω
L_{vfc}	Inductive droop coefficient of FC converter	2.31 H
R_{vlb}	Resistive droop coefficient of LB converter	0.2 Ω
R_{vsc}	Resistive droop coefficient of SC converter	0.83 Ω
C_{vsc}	Capacitive droop coefficient of SC converter	5.81 F

A. Parameter Tuning Guidelines

First, to investigate the impact of varying the parameters of the composite controller on the system's dynamic performance, a simulation study was conducted on a single bidirectional boost converter.

Since the composite controller consists of an observer and a controller, and the performance of the observer significantly influences the control performance of the controller, the control parameters of the observer should be tuned first. As shown in (18), the control parameters of the observer include k_{11i}, k_{12i}, k_{21i}, k_{22i}, k_{31i}, k_{32i}, α, and β. According to the results reported in [24], it can be assumed that $k_{11i} = k_{12i}$, $k_{21i} = k_{22i}$, and $k_{31i} = k_{32i}$. Thus, the parameters that actually require adjustment are only k_{11i}, k_{21i}, k_{31i}, α, and β. Initially, we roughly selected $k_{11i} = 200$, $k_{21i} = 30\,000$, $k_{31i} = 100$, $\alpha = 0.8$, and $\beta = 1.2$ based on a trial-and-error method. Fig. 4 presents the dynamic response of the observer's estimation of the total load under several typical scenarios (no-load to light load, light load to full load, and full load to medium load). It can be observed from the figure that decreasing k_{11i} yields a faster dynamic response but also increases overshoot, while increasing k_{21i} leads to a slower dynamic response but reduces overshoot. Under the typical operating conditions tested in this study, the sensitivity of the estimation results to variations in k_{31i} was observed to be low. However, it is emphasized that this parameter is a necessary component of the fixed-time observer framework

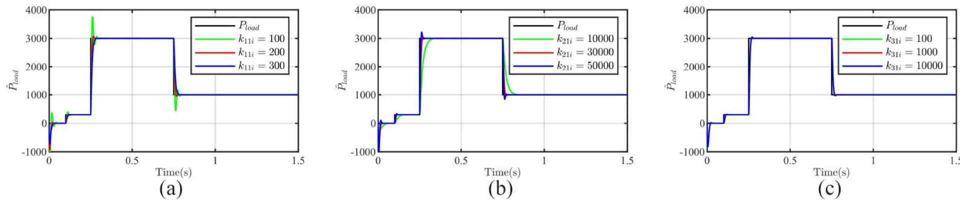

Fig. 4. Dynamic response of total load estimation with various FTSMDO parameters: (a) k_{11i}; (b) k_{21i}; and (c) k_{31i}.

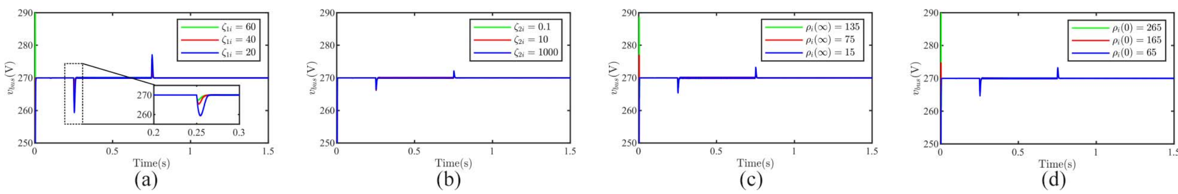

Fig. 5. Dynamic response of bus voltage with various PPC parameters: (a) ζ_{1i}; (b) ζ_{2i}; (c) $\rho_i(\infty)$; and (d) $\rho_i(0)$.

and is essential for theoretically guaranteeing the fixed-time convergence property.

Next, the control parameters of the PPC are tuned. Similarly, based on a trial-and-error approach, the parameters are roughly selected as $m_{1i} = 33$, $m_{2i} = 33$, $n_{1i} = 15$, $n_{2i} = 15$, $q_{1i} = 15$, $q_{2i} = 15$, $\zeta_{1i} = 40$, $\zeta_{2i} = 10$, $\rho_i(0) = 65$, $\rho_i(\infty) = 65$. Fig. 5 presents the influence of variations in the above parameters on the dynamic performance response of the converter's output voltage. It can be observed from the figure that as ζ_{1i} increases, the dynamic response becomes faster and the overshoot decreases, although the overshoot during startup increases. In contrast, changes in ζ_{2i} have no significant effect on the dynamic response. Moreover, reducing the initial and final values of the prescribed performance function (PPF) significantly reduces the overshoot during startup. Therefore, the PPC parameter tuning process should first adjust ζ_{1i} and ζ_{2i} to modify the dynamic performance under load variations after the system reaches steady state, and then tune the parameters of the PPF to adjust the dynamic performance during system startup.

Based on the above simulation analysis and Theorem 1, the parameter tuning procedure of the proposed composite controller is finally summarized as follows.

1) Step 1: Define the feasible range of all parameters according to stability and convergence theorems, to ensure that any parameter combination within this range can guarantee the fixed-time large-signal stability of the system:

For FTSMDO

a) Set $\alpha \in (0.8, 1)$, $\beta \in (1, 1.2)$ (consistent with the fixed-time convergence condition, with ($\varepsilon_1 = 0.2$, $\varepsilon_2 = 0.2$).
b) Set $k_{11i} = k_{12i}$, $k_{21i} = k_{22i}$, $k_{31i} = k_{32i}$ to reduce the dimension of tuning parameters.
c) Select k_{11i}, k_{21i}, k_{31i} to ensure the matrices A_i are Hurwitz, which is the necessary condition for the fixed-time convergence of the observer.

For PPC

a) Select odd integers m_{1i}, $m_{2i} > n_{1i}$, $n_{2i} > 0$, q_{1i}, $q_{2i} > p_{1i}$, $p_{2i} > 0$ (we recommend m/n = 11/5, p/q = 5/11, a widely used ratio in fixed-time control to avoid singularity and ensure convergence).
b) Determine PPF parameters $\rho(0)$ and $\rho(\infty)$ according to the physical safety constraints of the onboard system: $\rho(\infty)$ is set by the maximum allowable steady-state voltage error of the DC bus, and $\rho(0)$ is set by the maximum allowable transient voltage overshoot during startup.
c) Ensure control gains ζ_{1i}, ζ_{2i} satisfy $2\kappa_{1i} > \Theta$, $\kappa_{2i}2^{(p_{1i}+q_{1i}/2q_{1i})} > \Theta$, $2\kappa_{1i} > \bar{k}_{\xi_i}$, and $\kappa_{2i}2^{(p_{1i}+q_{1i}/2q_{1i})} > \bar{k}_{\xi_i}$ (derived from the Lyapunov stability proof in *Theorem 1*), which is the necessary condition for the fixed-time convergence of the closed-loop system.

2) Step 2: According to the sensitivity analysis results, the tuning order is determined by the degree of influence on observer performance:

a) Tune k_{11i} first: k_{11i} dominates the convergence speed of disturbance estimation. Start from the lower bound of the feasible domain, gradually increase k_{11i}, and select the value that achieves the fastest disturbance estimation convergence without obvious overshoot.
b) Then, tune k_{21i}: k_{21i} dominates the overshoot of disturbance estimation. Start from the upper bound of the feasible domain, gradually reduce k_{21i}, and optimize the overshoot on the premise of maintaining the convergence speed obtained in the previous step.
c) k_{31i} has no significant effect on the estimation performance, so it can be fixed to the median value of the feasible domain to avoid unnecessary tuning.

3) Step 3: According to the sensitivity analysis results, the tuning order is determined by the degree of influence on bus voltage performance:

a) Tune ζ_{1i} first: ζ_{1i} dominates the dynamic response speed and load transient overshoot of the bus voltage. Start from the lower bound of the feasible domain, gradually increase ζ_{1i}, and select the value that achieves the shortest settling time of the bus voltage under load step changes, while ensuring the startup overshoot is within the allowable range of the PPF.

b) Then, tune ζ_{2i}: ζ_{2i} has no significant effect on the main dynamic performance, so it can be fixed to the median value of the feasible domain, with fine-tuning only when there is residual steady-state error.

c) Finally, fine-tune $\rho(0)$ and $\rho(\infty)$: if the startup overshoot exceeds the allowable range, appropriately reduce $\rho(0)$ and $\rho(\infty)$ within the safety constraints to tighten the performance envelope.

B. System Simulation Verification Under Several Transient Conditions and Comparative Analysis

To validate the effectiveness and superiority of the designed controller, four test scenarios were established: 1) normal operation; 2) FC disconnected from HPSS during the takeoff and climb phase; 3) LB disconnected from HPSS during the takeoff and climb phase; and 4) SC disconnected from HPSS during the takeoff and climb phase. The load profile is as follows: no load from 0 to 3 s, a step change from no load to 300 W at 3 s (simulating the ground operation phase), an increase from 300 to 3000 W at 6 s (simulating the takeoff and climb phase), a decrease from 3000 to 1000 W at 12 s (simulating the cruise phase), an increase from 1000 to 3000 W at 15 s (simulating the maneuvering phase), and finally a step increase from 1000 to 8000 W at 20 s (simulating the activation of onboard equipment), The full characteristics of the load profile are presented in Fig. 6. In this scenario, the FC, LB, and SC autonomously supply power corresponding to low-frequency, medium-frequency, and high-frequency demand components, respectively. Throughout the entire process, the maximum bus voltage deviation was about 1.05 V. Fig. 7(b)–(d) presents the simulation results for the scenarios, in which the FC, LB, and SC were, respectively, disconnected from the HPSS during the takeoff and climb phase [with the same load profile as in Fig. 7(a)]. As can be observed from the figures, when one of the power sources is disconnected, the remaining sources autonomously compensate according to their respective response characteristics, supplying power within corresponding frequency bands. Throughout the entire process, the system remains stable, with a maximum bus voltage deviation of 1.9 V. The simulation results of the proposed FTSMDO's output under the above four scenarios are presented in Fig. 8, which demonstrates that the FTSMDO can rapidly and accurately estimate the load variations. Owing to the virtual impedance droop control implemented in this work, the FTSMDO actually estimates the load variations allocated to each converter via the power sharing strategy, that is, the low-frequency power components borne by the FC, the medium-frequency components by the LB, and the high-frequency components by the SC. For this reason, the dynamic performance of the FTSMDO cannot be directly visualized from the waveforms. Nevertheless, such dynamic performance can be indirectly reflected by the system-level simulation results shown in Fig. 7. The excellent dynamic performance presented in Fig. 7 fully verifies that the proposed FTSMDO can well meet the design requirements.

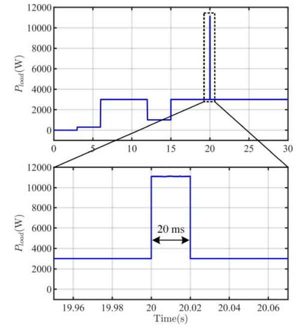

Fig. 6. Full characteristics of the load profile.

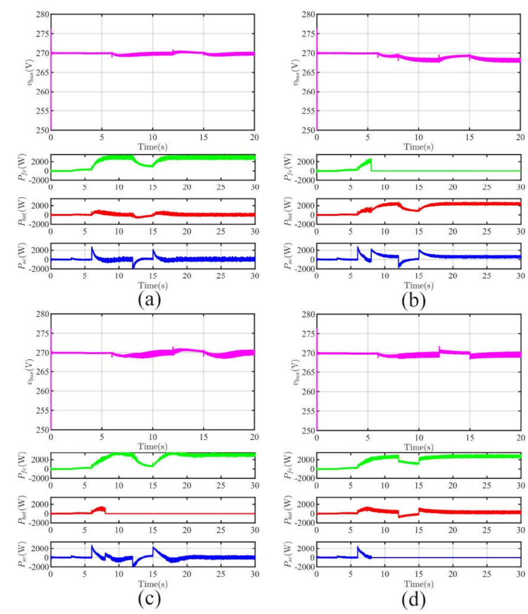

Fig. 7. Simulation results of bus voltage and power sharing in HPSS under four test scenarios: (a) normal operation; (b) FC disconnected from HPSS during the takeoff and climb phase; (c) LB disconnected from HPSS during the takeoff and climb phase; and (d) SC disconnected from HPSS during the takeoff and climb phase.

Figs. 9 and 10 present the simulation results of the DC bus voltage with variations in the filter capacitance and filter inductance of the FC, LB, and SC converters, respectively. It can be observed from Fig. 9 that when the filter capacitance is reduced from 1470 to 1000 μF, the bus voltage ripple increases slightly with almost no change in the dynamic response. In contrast, when the filter capacitance is increased from 1470 to 1940 μF, the bus voltage ripple is reduced, and the bus voltage fluctuation during the transient process is also mitigated. It can be observed from Fig. 10 that when the filter inductance is decreased from 0.6 to 0.42 mH, the bus voltage ripple and dynamic performance remain almost unchanged. Whereas when the filter inductance is increased from 0.6 to 0.78 mH, the bus voltage drop during load step-up is slightly improved, while the bus voltage overshoot during load step-down shows a marginal increase. It can be concluded from the above simulation results that

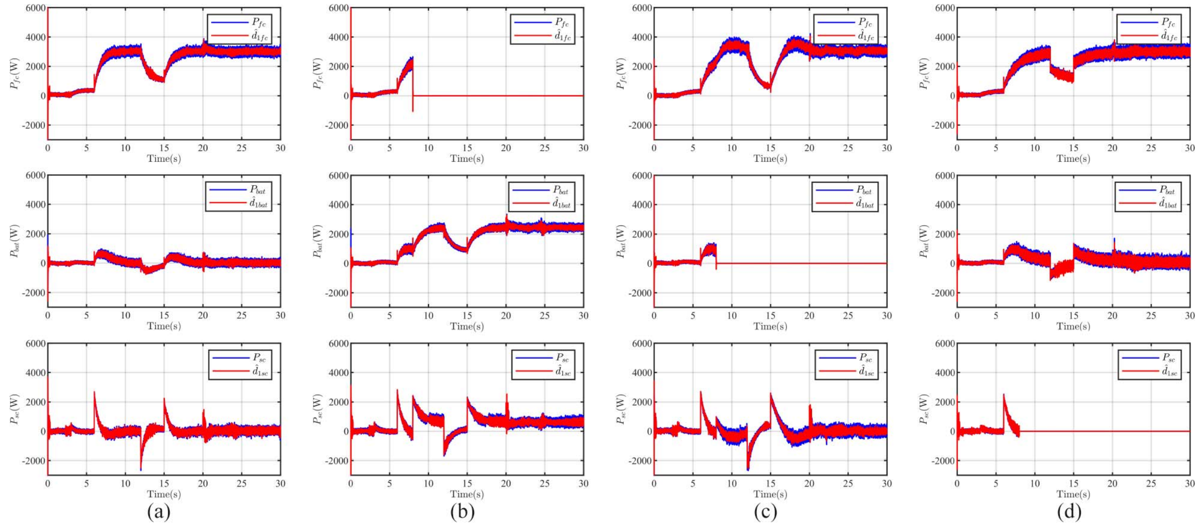

Fig. 8. Simulation results of FTSMDOs under four test scenarios: (a) normal operation; (b) FC disconnected from HPSS during the takeoff and climb phase; (c) LB disconnected from HPSS during the takeoff and climb phase; and (d) SC disconnected from HPSS during the takeoff and climb phase.

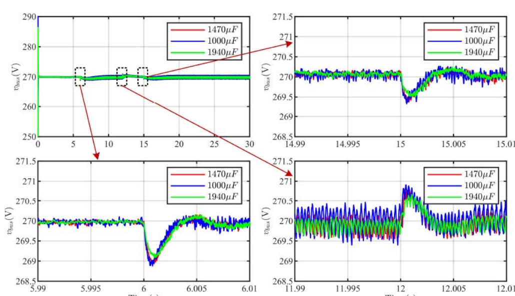

Fig. 9. Simulation results of the FC/LB/SC converter with filter capacitors of 1000, 1470, and 1940 μF.

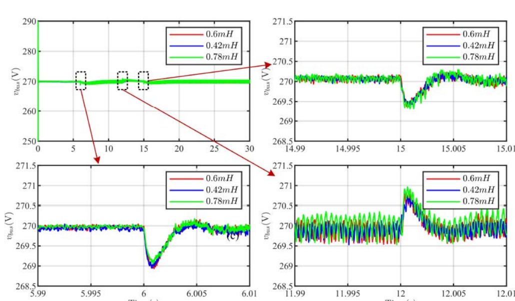

Fig. 10. Simulation results of the FC/LB/SC converter with filter inductors of 0.42, 0.6, and 0.78 mH.

the proposed composite controller exhibits strong robustness against parameter uncertainties.

Fig. 11 presents a comparison of simulation results of the dynamic response of the bus voltage between the proposed composite controller and a PI controller, under two scenarios: the process of load increase from 300 to 3000 W, and the process of FC disconnected from the HPSS while the load remains at 3000 W. As can be observed from the figure, the proposed

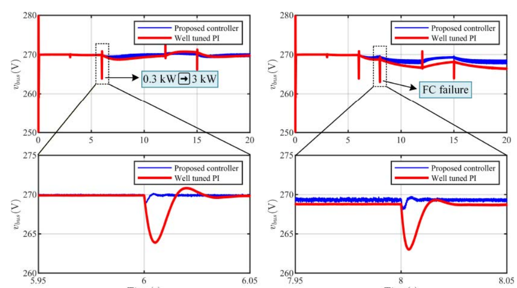

Fig. 11. Simulation results of bus voltage in HPSS under the proposed composite controller and PI controller.

composite controller exhibits a voltage deviation of 1.05 V and a settling time of 5 ms in both scenarios. In contrast, the PI controller shows a voltage deviation of 6.5 V and a settling time of 45 ms under the same conditions. For reference, another simulation results from the controller proposed in [16] under the same scenarios are shown in Fig. 12, which yields a voltage deviation of 1.1 V and a settling time of 18 ms. Therefore, while ensuring large-signal stability, the proposed controller achieves simultaneous improvement in both overshoot (voltage deviation) and settling time.

V. Experiment Results

To further validate the effectiveness of the proposed composite controller, a prototype experimental platform consistent with Fig. 1 was constructed in the laboratory, as shown in Fig. 13, on which the proposed composite controller was implemented.

A. Normal Operation

The dynamic response capability of the bus voltage and the power sharing performance of the HPSS under normal operation were first tested, as shown in Fig. 14. It can be observed

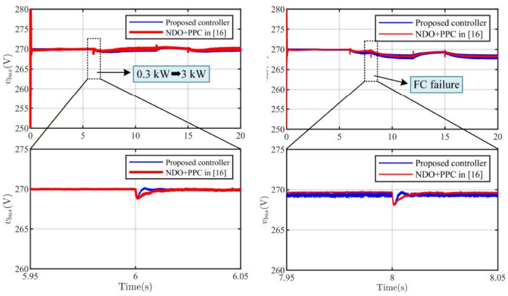

Fig. 12. Simulation results of bus voltage in HPSS under the proposed composite controller and controller in [16].

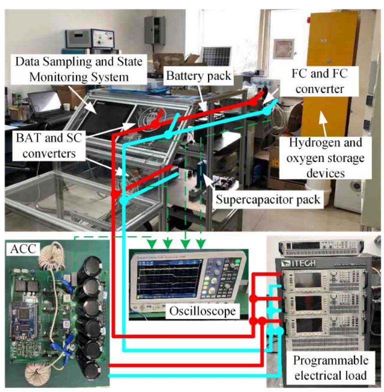

Fig. 13. Experiment platform.

from the figure that when the load increases from 300 to 3000 W, under the PI controller [Fig. 14(a)], the regulation time and the maximum voltage deviation of the bus voltage are about 20 ms and 7 V, respectively. In contrast, under the proposed composite controller [Fig. 14(b)], the regulation time and the maximum voltage deviation are about 5 ms and 2.5 V, respectively, which effectively verifies the effectiveness and superiority of the proposed composite controller. Moreover, it can also be seen from the figure that during the load variation, the FC, LB, and SC autonomously provide low-frequency, medium-frequency, and high-frequency power, respectively, validating the efficacy of the power sharing controller.

B. Partial Power Sources Disconnected

Subsequently, we tested the dynamic response capability of the bus voltage and the power sharing performance when the FC, LB, and SC were individually disconnected from the HPSS, as shown in Figs. 15–17. It can be observed from the figures that under the PI controller, the voltage deviation is largest (7 V) when the FC is disconnected under a 3000 W load, whereas under the proposed composite controller, the voltage deviation for the same scenario is 3 V. Under the same conditions, the disconnection of the LB and SC has almost no impact on the bus voltage, as the LB and SC do not provide power in this operating state. Furthermore, after the individual disconnection of the FC, LB, and SC, when the load increases from 1000 to 3000 W, the maximum voltage deviation under the PI controller is 9 V,

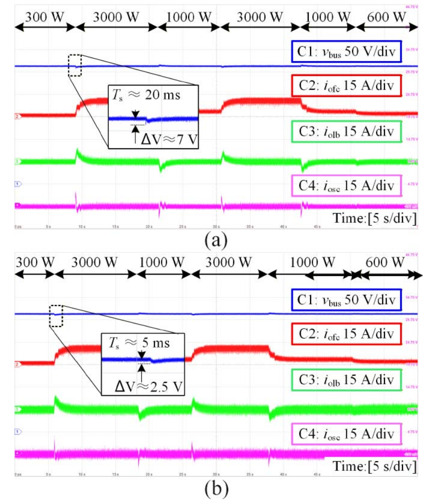

Fig. 14. Dynamic response of bus voltage and power sharing of the HPSS under normal operation with different controllers: (a) PI controller; and (b) proposed controller.

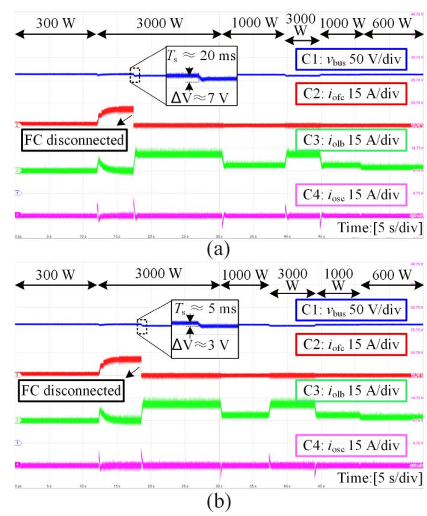

Fig. 15. Dynamic response of bus voltage and power sharing of the HPSS when FC is disconnected with different controllers: (a) PI controller; and (b) proposed controller.

compared to 4.5 V under the proposed composite controller. The bus voltage regulation time shows no significant change compared to normal operation. These results effectively validate the superiority of the proposed composite controller.

C. Parameter Variation

Finally, the performance of the power supply system under parameter variations was tested, specifically the performance with the decrease and increase of the filter capacitance and filter inductance of the FC, LB, and SC converters. The corresponding experimental results are presented in Figs. 18 and 19. It can be observed from Fig. 18(a) that when the filter capacitance is reduced from 1470 to 1000 μF, the dynamic process of the DC bus voltage remains almost unchanged. It can be observed from Fig. 18(b) that when the filter capacitance is

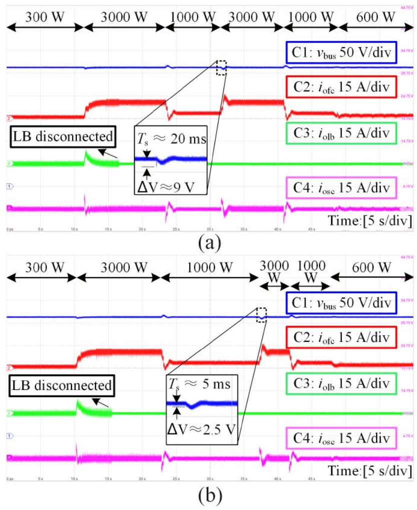

Fig. 16. Dynamic response of bus voltage and power sharing of the HPSS when LB is disconnected with different controllers: (a) PI controller; and (b) proposed controller.

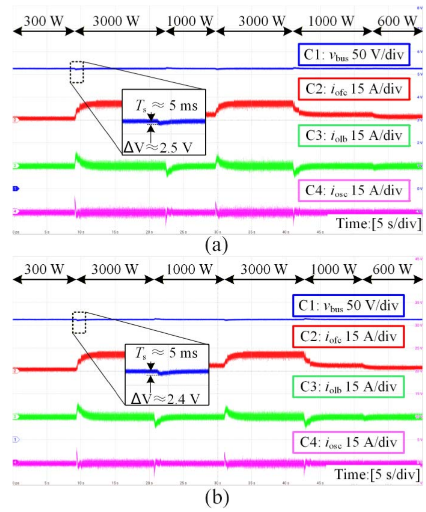

Fig. 18. Dynamic response of bus voltage and power sharing of the HPSS under filter capacitance variation: (a) $C_{\rm fc} = C_{\rm lb} = C_{\rm sc} = 1000$ μF; and (b) $C_{\rm fc} = C_{\rm lb} = C_{\rm sc} = 1940$ μF.

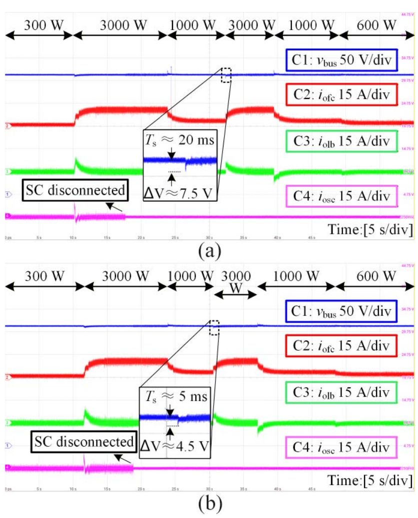

Fig. 17. Dynamic response of bus voltage and power sharing of the HPSS when SC is disconnected with different controllers: (a) PI controller; and (b) proposed controller.

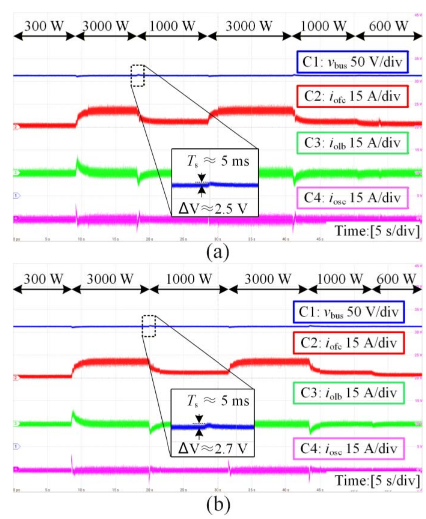

Fig. 19. Dynamic response of bus voltage and power sharing of the HPSS under filter inductance variation: (a) $L_{\rm fc} = L_{\rm lb} = L_{\rm sc} = 0.42$ mH; and (b) $L_{\rm fc} = L_{\rm lb} = L_{\rm sc} = 0.78$ mH.

increased from 1000 to 1940 μF, the bus voltage fluctuation is slightly mitigated, while the dynamic response speed shows no variation. It can be observed from Fig. 19(a) that when the filter inductance is decreased from 0.6 to 0.42 mH, the bus voltage ripple and dynamic performance remain almost unchanged. It can be observed from Fig. 19(b) that when the filter inductance is increased from 0.6 to 0.78 mH, the bus voltage drop during load step-up is slightly improved, whereas the bus voltage overshoot during load step-down shows a marginal increase. These experimental results are consistent with the simulation results shown in Figs. 9 and 10, which further sufficiently validates the effectiveness of the proposed composite controller against parameter uncertainties.

Based on the experimental results presented above, it can be concluded that the proposed composite controller not only ensures the large-signal stability of the HPSS but also achieves superior dynamic performance and strong robustness against parameter uncertainties, which aligns with the theoretical analysis and simulation results.

VI. Conclusion

This article developed a composite controller by integrating an FTSMDO and a PPC to address voltage degradation and instability in onboard HPSSs under various transient events. A systematic parameter tuning procedure was provided, supported by rigorous theoretical analysis. Simulation and experimental results demonstrate that the proposed controller enhances the system's dynamic performance while ensuring large-signal stability, maintaining prescribed performance during critical transients such as wide-range load switching and partial source disconnection. This effectively guarantees the secure and stable operation of the hybrid power system, supporting highly reliable AAV operation. Furthermore, the proposed composite controller can also be extended to the design of controllers for converters with multiple topological structures.

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